Integrated Industry Synthesis: A Governed Question Lifecycle for HKDSE ICT Paper 1A

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Overview and Industry Context

Hong Kong secondary students preparing for the HKDSE Information and Communication Technology (ICT) examination practise on a finite resource. Paper 1A is a 40-question multiple-choice paper, and the 2012–2025 papers give a bank of 560 questions. Past papers cannot be re-issued, yet students need targeted practice on specific topics. In earlier capstones I built a tutor that served this bank and a generator that could extend it, and together they exposed four gaps. The usable bank is small and uneven: Module A has 154 questions, Module E has 50, and one sub-topic has only 3. Generation was untrustworthy, since only 1 of 10 hand-read TinyLlama samples was usable without editing. Difficulty exists only as a proxy of past cohort performance. And safeguards stopped at the tool level, so a fluent but wrong final reply could still reach a student.

I framed the project as one question: can classified and generated items be added to an adaptive tutor's bank safely, when each weakness is addressed by a measurable check and a human decision, with difficulty left honestly unknown? Education suits this problem because the data is structured and no single technique covers all of it. The answer I built is a governed lifecycle in which AI proposes, statistics and classifiers check, a teacher decides, and the tutor serves only what has been admitted.

Integration Rationale

The system integrates four earlier capstones, each handling a different part of the problem. Capstone 6 (agentic AI) supplies the delivery layer, a tutor that selects, grades, explains and tracks progress. It also supplies the demand signal, because its thin question pools show where supply is missing. Capstone 5 (generative AI) supplies candidate questions, and its

documented failures (runaway output, overlapping distractors, no answer key, answers given away) became the requirements for the verification gate. Capstone 3 (machine learning) supplies a text classifier that labels the 2025 and other unclassified questions. I reused its workflow of comparing model families on one metric and tuning the winner, while the TF-IDF features, temporal split and abstention are new. Capstone 2 (statistics) supplies gap detection, Welch's t-test, effect sizes and confidence intervals, so decisions rest on measurement rather than judgement by eye. Capstone 4 (deep learning) is excluded by design: figure-aware generation is out of scope, so an image classifier had no natural role.

The components form a chain rather than a collection. Labels are proposed and teacher-approved, gaps are detected, candidates are generated for those gaps, the gate checks them, a teacher reviews them, approved items are admitted as provisional, and the tutor delivers them. Each stage consumes the previous stage's output. Reusing one domain and one bank across Capstones 5, 6 and 7 was deliberate, because a single coherent story is stronger evidence of intentional design than three unrelated pieces.

System Design and Technical Decisions

The system has a staff side and a student side. On the staff side, the official question file stays immutable, and all changing state lives in a governed SQLite store: label proposals, candidates, gate results, review decisions and admitted items. The student-side tutor opens that store read-only and has no tool that can generate or admit an item.

Several choices follow from earlier failures. The classifier is a logistic regression on TF-IDF features, selected by macro-F1 over folds grouped by shared-context cluster. It is evaluated on a temporal split (train 2012–2022, test 2023–2024) because the real task is labelling an unseen year. It abstains below a confidence of 0.50 rather than forcing a label, and proposes a sub-topic

only where at least 15 training questions exist. Gap detection marks a sub-topic pool below 15 units as a gap. The generator is backend-agnostic, driven by one configuration entry per model, which let me run a bake-off of the re-created Capstone 5 TinyLlama adapter, Gemma 4 E4B and Claude Haiku 4.5 on six thin sub-topics, producing 54 candidates.

The gate has six deterministic checks and calls no model: format and length, figure and dangling references, duplicate or overlapping options, copy ratio against real questions, key length, and stem–option overlap. These parallel established item-writing guidance on overlapping options and clues to the answer (Haladyna et al., 2002). Thresholds were calibrated on 400 real questions, so each rule is judged by how many genuine exam items it would wrongly flag. A seventh, batch-level check uses statistics to compare candidate length with real questions. I kept a model out of the gate so that the control itself behaves identically on every run and can be tested.

The tutor uses a hybrid router: deterministic rules first, then a bounded language-model Orchestrator with a step limit and one Explainer worker. This follows the advice to use the simplest solution and add agency only where flexibility justifies the cost (Anthropic, 2024). Labels, answer keys and verdicts are added by code rather than by model wording, and a failed or inconsistent explanation falls back to a fixed message. The cost is that the Orchestrator sends messages to an external service and is not deterministic.

Ethical Considerations and Responsible AI

The central ethical question is who is accountable when an AI-drafted exam item reaches a student. A model cannot own the correctness of an item, so a named teacher approves every label and every admitted item. In this work that meant 48 label reviews and 46 candidate decisions, producing 42 admitted items. The NIST framework treats accountability and

documented human oversight as core parts of governing AI risk (Tabassi, 2023). The tutor's read-only access means no route to a student bypasses that decision.

On transparency, admitted items are provisional, carry no difficulty value, and are always shown with the label "AI-drafted, teacher-approved, not an HKEAA question", added by code. Inventing a difficulty estimate would present a guess as a measurement. Documenting a system's intended use and limits is the spirit of model cards (Mitchell et al., 2019), and it is why my evaluation states what each component does not establish.

On bias and equity, the curriculum is imbalanced, thin sub-topics and figure- or table-dependent questions are under-served, and classifier quality may differ by module. I reported per-module and per-year results with confidence intervals and chose abstention over forced labels. The teacher's blind sample also revealed that 9 of 30 questions, all from before 2025, test word-processing or presentation software no longer in the syllabus. Students could meet such items, so I report this as a finding rather than hide it.

Other risks remain. The questions are HKEAA copyright, and whether my use is fair dealing is a legal question this project does not settle. In Model mode the tutor sends student text to an external service. Rules mode is the default, no real student data was used, and deployment would require consent under Hong Kong's Personal Data (Privacy) Ordinance. Finally, one teacher made every decision, so nothing shows how a second teacher would decide.

Evaluation and Reflection

The system answers its central question under the conditions I built, and not beyond them. Of 54 candidates, the gate passed 29, flagged 17 and rejected 8. The teacher approved all 29 passed items and 13 of the 17 flagged ones. All four teacher rejections came from the flagged group, so 42 of 46 reviewed candidates (91.3%) were admitted. The teacher's answer matched

the generator's key in 30 of 32 keyed items, whereas TinyLlama wrote no key at all. The classifier answered 59 of 75 held-out questions and was right on 57 (96.6%), against 85.3% for the Capstone 5 proxy, which answered every question. Counting abstentions as errors, the classifier scored 76.0%. The tutor passed all 14 scripted scenarios, and the Orchestrator chose the intended action on 12 of 12 plain-wording messages against 7 of 12 for the rules.

Because the samples are small, I put Wilson confidence intervals on these shares (Brown et al., 2001). The overall approval rate has a 95% interval of 0.797 to 0.966, and the confidence intervals for the three backends overlap, so I do not rank them. A statistical check also showed that Haiku's stems were much longer than real questions (mean 306 against 186 characters), which a student could use to spot AI items.

Several things did not work. The rule router missed 5 of 12 plain-wording messages. The 8 gate rejects were never reviewed, so the gate's rate of wrongly rejecting good items is unknown. The Explainer's word limit is requested in its prompt, not enforced in code. One teacher made every decision, each backend was sampled once, and no students used the items, so I claim nothing about learning or about the quality of distractors. The 254 passing tests check logic, not live model behaviour.

Four lessons came from integrating across domains. First, I kept revisiting the data. Every generative and agentic step sent me back to examine the question bank again, from thresholds calibrated on real items to the blind sample that exposed the old-syllabus questions. Second, a language-model agent is easy to lose control of, so in a system like this one, rules should own the state, the labels and the safety checks, and the model should only read plain words within a step limit. My own results show the balance: the rules alone missed 5 of 12 messages, but the Orchestrator still needed code around it, and the Explainer's word limit, which I only requested

in a prompt, was not a rule at all. Third, the statistical foundation of Capstone 2 kept my decisions evidence-based and kept my claims honest about small samples, for example when the confidence intervals overlapped and I declined to rank the backends. Fourth, protection for students comes from the gate, the teacher's decision and the tutor's read-only access working together. The gate stops known failures early and spares the teacher's time, so that only admitted content reaches students.

Professional Relevance

This project shows the work of an applied AI role rather than model building alone. I chose methods to fit the problem, combined statistics, machine learning, generation and agents in one pipeline, and designed governance in from the start: a named approver, a record of every decision, read-only access for the student side, and labels applied by code. I also evaluated honestly, with confidence intervals on small samples, failure cases reported, and the limits of each result stated. The work is reproducible: components are importable modules, and 254 tests pass in a fresh environment built from a short requirements file.

For real use, the same approach implies steps I would insist on: a second reviewer to measure agreement, a pilot with students before any claim about learning, consent and data minimisation before any model sees student text, and legal clarity on the use of examination material. Those are the conditions under which I would stand behind this system in an education technology setting.

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